FORE SCHOOL OF MANAGEMENT



Course Name: Agentic AI for Business Automation

Project Report

Infosys Annual Report Intelligence

A Multi-Year Financial Analysis Using Retrieval-Augmented Generation and Agentic AI

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1. Executive Summary

The project develops a document-grounded financial analysis workflow for Infosys using five annual reports, FY2022 to FY2026. It demonstrates how long-form corporate disclosures can be converted into a searchable evidence base and used to support a structured, multi-year review. The workflow includes PDF ingestion, page-aware text extraction, chunking, vector indexing, semantic retrieval, financial comparison and an agent-based synthesis stage.

The implementation uses Python and pypdf for text extraction; ChromaDB for persistent vector storage; Ollama to serve local models; nomic-embed-text for embeddings; qwen2.5 for answer generation; and crewAI to coordinate analyst tasks. The source code is modular, with separate scripts for extraction, retrieval, financial-data preparation, analysis and retrieval testing. The key outputs are a Markdown analysis and an SVG trend chart.

The five-year comparison shows revenue rising from ₹1,21,641 crore in FY2022 to ₹1,78,650 crore in FY2026, approximately 46.9% growth. Net profit increased from ₹22,110 crore to ₹29,440 crore, approximately 33.2%. Operating margin was around 21% in the later years, with the FY2026 figure explicitly labelled adjusted. Free cash flow peaked at ₹34,549 crore in FY2025 before easing to ₹33,097 crore in FY2026.

The project demonstrates that a useful business-analysis assistant needs more than a language model. Document quality, chunk boundaries, metadata, retrieval relevance, source traceability and numerical validation all influence the result. The system is intended as a decision-support prototype: the annual reports remain the authoritative evidence, and users should verify material figures and conclusions against the original pages.

GITHUB REPOSITORY

2. Project Background and Business Context

Annual reports are central sources for understanding a listed company's performance, strategy, risks and governance. They combine narrative discussion with financial statements, notes, tables and year-specific explanations. Reviewing multiple reports manually can be time-consuming because an analyst must locate the same indicators across different years, interpret definitions and adjustments, and then assemble a comparable view.

Infosys was selected as the subject because its annual reports provide a consistent multi-year corporate disclosure corpus suitable for demonstrating document retrieval and business analysis. The project uses FY2022–FY2026 reports and focuses on selected indicators rather than attempting to analyse every business line, disclosure or accounting note.

The business use case is a financial analyst, management student or business decision-maker asking a natural-language question such as "What was consolidated revenue in FY2024?" or "How did free cash flow change between FY2025 and FY2026?" The system aims to retrieve relevant evidence quickly, make the source page identifiable and support a structured comparison.

Retrieval-Augmented Generation (RAG) connects a language model to an external document collection. Rather than depending only on information learned during model training, the system searches the selected corpus and provides relevant passages as context for the answer. An agentic workflow adds task coordination, allowing numerical analysis and strategic interpretation to be handled as distinct stages before synthesis.

3. Problem Statement and Objectives

Problem statement: How can a local, reproducible document-analysis workflow make five years of Infosys annual-report information searchable, retrieve passages relevant to natural-language questions, and organise selected financial indicators into a traceable multi-year review? The project was guided by the following objectives:

- Extract the five annual reports into page-aware text files.
- Create a persistent semantic index for natural-language retrieval.
- Retain source metadata so passages can be traced to a financial year and PDF page.
- Compare selected financial indicators over five years.
- Use sequential agent tasks to organise financial analysis and risk/strategy interpretation.
- Validate a representative retrieval question and document limitations.

4. Scope, Assumptions and Source Documents

The source corpus consists of the Infosys annual reports for FY2022, FY2023, FY2024, FY2025 and FY2026. The PDFs are retained as source documents under data/raw; their extracted text representations are stored under data/processed. The project's final financial analysis focuses on revenue, net profit, operating margin and free cash flow, with year-on-year growth calculated for revenue and net profit.

Official source links are listed below so that a reviewer can consult the company's own investor-relations material. The public GitHub repository contains the code and selected output artifacts; large PDFs and the vector database are best retained in a separate backup rather than unnecessarily duplicated in the repository.

- **Infosys Annual Reports archive:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports.html
- **FY2023 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2022-23.html
- **FY2024 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2023-24.html
- **FY2025 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2024-25.html
- **FY2026 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2025-26.html

Scope exclusions: the project does not perform a full valuation, forecast share prices, provide investment advice, benchmark Infosys against peers, or exhaustively analyse every qualitative disclosure. It does not include OCR for image-only pages. The financial figures below reflect the project's checked output and should be reconciled with the exact definitions and notes in each source report before formal use.

5. Data Preparation and PDF Text Extraction

The first stage is document ingestion. Five PDF files are stored separately from the processed text, preserving the source documents for verification. The extraction script src/extract_reports.py uses pypdf to read the PDFs page by page. Page markers are inserted into the extracted text so a passage can be traced to its original PDF page.

The observed PDF page counts were FY2022: 363, FY2023: 359, FY2024: 353, FY2025: 370 and FY2026: 384. A small number of pages yielded no extractable text: FY2022 page 362, FY2023 page 358 and FY2026 page 2. The current implementation does not run OCR, so image-only content on those pages may not be searchable.

Page-aware extraction is important because an answer is more useful when it includes source context. A retrieved paragraph can be checked against nearby headings, footnotes and table structure in the original report. Metadata also helps distinguish the same indicator across different reporting years.

After extraction, the corpus is split into chunks for semantic indexing. Chunking balances two competing needs: small chunks are focused and easier to retrieve, while larger chunks preserve context such as headings, table labels and surrounding explanation. If chunks are too small, a financial figure may become separated from its row label; if they are too large, retrieval can return unnecessary material.

6. System Architecture and Technology Stack

The project is implemented as a sequence of small components, rather than one monolithic script. This supports debugging, testing and re-running a particular stage without repeating every step. The indexing pipeline is run before question answering; at question time, the query is embedded and matched against stored document vectors.

Architecture flow: Infosys annual-report PDFs → pypdf extraction → page-aware text → chunking and metadata → nomic-embed-text embeddings → ChromaDB persistent index → user query and semantic retrieval → qwen2.5 response generation → crewAI analyst workflow → Markdown report and SVG chart.

The architecture can be divided into three layers. The data preparation layer handles PDFs, extracted text and chunks. The retrieval layer handles embeddings, vector storage, similarity search and evidence assembly. The analysis layer handles financial calculations, agent task coordination and report output.

Python is used for application logic and orchestration. pypdf extracts text. Ollama serves the local models. nomic-embed-text represents text as vectors. ChromaDB stores vectors and source metadata. qwen2.5 generates natural-language responses from retrieved context. crewAI coordinates the sequential analyst roles. Markdown and SVG provide portable output formats, while GitHub supports version control and sharing of the source code.

7. RAG Engine: Indexing, Retrieval and Grounding

Indexing begins by reading the extracted annual-report text, splitting it into manageable chunks and attaching metadata such as financial year, source filename, PDF page and chunk number. The embedding model converts each chunk into a numerical vector that represents semantic content. ChromaDB persists the vectors and associated text so that later queries do not need to rebuild the index each time.

At question time, the query is embedded using the same embedding model. The vector database searches for chunks that are semantically close to the question. The selected passages are assembled as context and passed to qwen2.5. The generated response should use the retrieved evidence and preserve source details so that a user can inspect the original annual report.

A functional retrieval check was performed with the question “Infosys consolidated revenue in FY2024”. The retrieved passage returned ₹1,53,670 crore and identified FY2024.pdf, PDF page 318. This confirms that the pipeline can connect a natural-language question to relevant report evidence and a specific source location. It should not be treated as a full benchmark of accuracy across all possible questions.

Grounding is the central quality control. A fluent answer can still be wrong if the wrong passage is retrieved, a table is misread, or the language model confuses values from adjacent years. The project therefore retains source metadata and includes a dedicated retrieval test. For high-stakes use, every important number should be compared with the original report and relevant footnotes.

8. Agentic Workflow and Report Generation

The agentic stage separates the analysis into roles with different responsibilities. The Financial Analyst task focuses on the financial indicators, numerical movement and multi-year comparisons. The Risk & Strategy Analyst task reviews business implications and relevant disclosures, while distinguishing reported facts from interpretations. A synthesis task brings the outputs together into a coherent report.

The sequential design helps keep numerical analysis and interpretation separate. For example, a decline in free cash flow is an observed trend; its cause cannot be inferred from the trend alone. A follow-up review would need to examine operating cash flow, working-capital movements, capital expenditure and the report’s notes. Similarly, stable operating margin is an observation, while any explanation of why it remained stable needs additional evidence.

Agent orchestration does not itself guarantee factual accuracy. Tasks can pass an unsupported assumption from one stage to another, and a polished final summary can make uncertain reasoning sound definitive. The workflow should therefore be treated as an organising mechanism, with retrieval, source verification and human review acting as quality controls.

The report-generation output is saved in Markdown, which is easy to inspect, revise and version-control. The chart is stored as SVG so it can scale cleanly when embedded in a report or presentation. The public repository includes the code and selected outputs, making the project structure inspectable.

9. Financial Analysis and Results

The following tables present the four indicators used in the project’s five-year comparison. Amounts are expressed in ₹ crore. FY2026 operating margin is marked adjusted in the project output, and that qualification should remain visible wherever the number is reproduced.

Revenue rose from ₹1,21,641 crore in FY2022 to ₹1,78,650 crore in FY2026, an increase of approximately 46.9%. Year-on-year growth was 20.66% in FY2023, 4.70% in FY2024, 6.06% in FY2025 and 9.61% in FY2026. The series shows that revenue increased throughout the period, although the growth rate varied materially by year.

Net profit increased from ₹22,110 crore in FY2022 to ₹29,440 crore in FY2026, approximately 33.2%. Growth was 8.98% in FY2023, 8.87% in FY2024, 1.83% in FY2025 and 10.21% in FY2026. The FY2025 result is a reminder that revenue growth and net-profit growth do not necessarily move at the same pace.

Operating margin was 23.0% in FY2022, 21.0% in FY2023, 20.7% in FY2024, 21.1% in FY2025 and 21.0% adjusted in FY2026. The values are clustered around 21% from FY2023 onward, but the adjusted label means the series should not be interpreted as perfectly like-for-like without consulting each annual report's definition.

Free cash flow was ₹22,803 crore in FY2022, ₹20,443 crore in FY2023, ₹23,865 crore in FY2024, ₹34,549 crore in FY2025 and ₹33,097 crore in FY2026. It reached its highest value in FY2025 and then declined approximately 4.2% in FY2026. The change is an observation, not a conclusion about the cause or the company's financial health.

Table 1. Five-Year Financial Comparison

Financial year	Revenue (₹ crore)	Net profit (₹ crore)	Operating margin	Free cash flow (₹ crore)
FY2022	1,21,641	22,110	23.0%	22,803
FY2023	1,46,767	24,095	21.0%	20,443
FY2024	1,53,670	26,233	20.7%	23,865
FY2025	1,62,990	26,713	21.1%	34,549
FY2026	1,78,650	29,440	21.0% (adjusted)	33,097

Table 2. Year-on-Year Growth

Financial year	Revenue growth YoY	Net profit growth YoY
FY2023	20.66%	8.98%
FY2024	4.70%	8.87%
FY2025	6.06%	1.83%
FY2026	9.61%	10.21%

10. Business Implications

- **Growth monitoring:** the five-year increase in revenue is positive in direction, but the variation in year-on-year growth means that a decision-maker should look beyond absolute revenue. Segment and geography disclosures, client demand and the management discussion may help explain the changing pace.
- **Revenue-to-profit conversion:** net profit increased less than revenue over the full period. This suggests that costs, taxes, exceptional items and margin movement are relevant areas for further analysis. The trend alone does not establish which factor drove the difference.
- **Margin comparability:** operating-margin values should be interpreted using the company's definitions and adjustments. Reports and presentations should label adjusted figures clearly and avoid implying exact comparability where definitions differ.
- **Cash-flow resilience:** free cash flow was lower in FY2026 than FY2025. A useful next step is to review operating cash flow, capital expenditure and working-capital movements. One year of decline is not sufficient evidence of a persistent problem.
- **Decision support:** the strongest practical benefit of the system is the ability to locate relevant source passages quickly and structure comparisons consistently. It can help analysts focus their time on interpretation and verification, but it does not replace professional judgement.

11. Evaluation, Validation and Quality Controls

Evaluation was conducted at prototype level. The retrieval test for FY2024 consolidated revenue returned the expected figure and an identifiable source page. The codebase also contains a test_retrieval.py script so retrieval can be checked separately from the report-generation workflow.

The validation approach checks three distinct layers: (1) whether the relevant passage is retrieved, (2) whether the generated response accurately reflects the passage, and (3) whether the final numerical comparison is consistent with the figures used in the analysis. These layers should not be collapsed into one pass/fail judgement because a correct-looking answer may still have weak source support.

Table 3 summarises the validation evidence available for this submission. The single confirmed question is reported as a functional test, not as a statistical accuracy score. A larger evaluation set should contain questions, expected values, exact source pages and pass/fail criteria, including cases involving footnotes and tables.

Recommended quality controls are: check the original page for every material number; preserve source filename and page metadata; explicitly label adjusted values; distinguish disclosed risks from analyst interpretation; document extraction gaps; and rerun retrieval tests after changing chunking, embeddings or models.

Test / control	Evidence available	Interpretation
FY2024 revenue query	Retrieved ₹1,53,670 crore from FY2024.pdf, PDF p.318	Functional retrieval check passed
Page traceability	Year, filename and PDF page metadata retained	Supports source inspection
Financial trend review	Five-year indicator table and YoY calculations	Numerical series can be reviewed separately
Limitations log	Extraction gaps and adjusted margin noted	Prevents overclaiming completeness

Table 4. Problems and Responses

Problem	Impact	Response / improvement
A few pages had no extractable text	Possible gaps in searchable corpus	Record gaps; add OCR in future
Chunk boundaries may split table context	Potential misinterpretation of figures	Check retrieved passage and source page
Local model/environment dependencies	Setup may differ across machines	Document dependencies and model requirements
Git identity initially unconfigured	Commit could not be created	Set username/email and commit source
Large PDFs/index unsuitable for lightweight repo	Repository size and portability concerns	Use GitHub for code and separate archive for full data

12. Problems Faced and Solutions

PDF text extraction does not always produce complete text. A few pages yielded no extractable content, so the project records the limitation rather than assuming that every page is searchable. OCR is a future enhancement for scanned or image-based pages.

Financial tables and narrative may be split across chunks. If a value becomes separated from its row label or footnote, the system can retrieve relevant-looking but incomplete context. Page metadata and inspection of retrieved passages help identify such cases; table-aware extraction would improve reliability further.

Local model availability and environment setup affect reproducibility. The workflow depends on a compatible Python environment, installed dependencies, Ollama and the required local models. The README and requirements file help document setup, but a clean-machine reproduction test remains a useful next step.

Git initially could not create a commit because user identity was not configured. The issue was resolved by configuring a Git username and email, then committing the source files. The repository was subsequently published to GitHub.

Large generated artifacts such as the ChromaDB index and annual-report PDFs are not ideal for a lightweight public Git repository. The project keeps code and selected outputs in GitHub and recommends a separate archive for the full working data, which reduces repository size and makes the source code easier to browse.

13. Limitations, Risks and Future Enhancements

- **Coverage limitation:** the current analysis covers four indicators and five reporting years. It is not a complete assessment of Infosys's financial position, operating segments, governance, strategy or market valuation.
- **Extraction limitation:** PDF text extraction can miss content embedded as images or poorly encoded tables. OCR and more robust table extraction would improve coverage.
- **Retrieval limitation:** semantic similarity is a ranking signal, not proof that a passage fully answers a question. Questions requiring calculations across several pages may need multi-step retrieval and explicit computation.
- **Model limitation:** language models may misstate figures, blend years, omit qualifications or generate causal explanations unsupported by the evidence. Source review remains essential.
- **Reproducibility limitation:** rebuilding the full system requires access to the source PDFs, Python packages, Ollama models and adequate local compute. The persistent ChromaDB index can be backed up, but should be regenerated if its source corpus or embedding configuration changes.

Future enhancements include OCR, structured extraction of financial tables, more financial indicators, peer benchmarking, interactive dashboards, automated citation validation and a larger question-answer test set. A production version would also require stronger monitoring, version control for the source corpus and clearer user-facing uncertainty handling.

14. Conclusion

The project demonstrates a complete document-to-analysis workflow for Infosys annual reports from FY2022 to FY2026. It combines page-aware extraction, semantic indexing, source-linked retrieval, financial comparison and sequential agent-based synthesis. The output provides a consistent starting point for reviewing revenue, net profit, operating margin and free cash flow across five years.

The financial comparison shows that revenue and net profit increased over the period, operating margin remained around 21% in the later years, and free cash flow peaked in FY2025 before declining in FY2026. These patterns support further questions rather than definitive causal conclusions. The annual reports remain the authoritative source for interpretation.

The main project learning is that reliable business automation depends on the quality of the complete pipeline: source documents, extraction, chunking, metadata, retrieval, calculations, synthesis and verification. The modular implementation provides a foundation for further enhancements while keeping the limitations visible.

References

1. **NSE India** https://www.nseindia.com/companies-listing/corporate-filings-annual-reports
2. **Infosys Annual Reports archive:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports.html
3. **FY2023 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2022-23.html
4. **FY2024 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2023-24.html
5. **FY2025 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2024-25.html
6. **FY2026 Integrated Annual Report:** https://www.infosys.com/investors/reports-filings/annual-report/annual-reports/ar-2025-26.html
7. **Project source code and selected outputs:** https://github.com/Ishita-fore/infosys-annual-report-intelligence

Appendix A. Repository Structure and Deliverables

```
infosys-annual-report-intelligence/
├── .gitignore
├── README.md
├── requirements.txt
├── src/
```

```
│   ├── extract_reports.py
│   ├── rag.py
│   ├── financial_data.py
│   ├── analyze_reports.py
│   └── test_retrieval.py
└── outputs/
    ├── infosys_five_year_analysis.md
    └── financial_trends.svg

Local working data (kept separately from the lightweight repository):
data/
├── raw/        # FY2022.pdf ... FY2026.pdf
├── processed/  # FY2022.txt ... FY2026.txt
└── chroma_db/  # persistent vector index
```

Appendix C. Representative Questions for Further Evaluation

The following question set can be used to extend the current prototype evaluation. Only the FY2024 revenue question was confirmed in the recorded functional test; the remaining questions are proposed test cases and should be run and checked against the original reports before being described as passed.

No.	Proposed question	Evidence to verify	Evaluation focus
1	What was consolidated revenue in FY2024?	FY2024 report, PDF p.318 (retrieved in the recorded test)	Correct year, figure and source location
2	How did revenue change from FY2025 to FY2026?	Revenue figures and management discussion in both reports	Correct subtraction and growth calculation
3	What was net profit in FY2025?	FY2025 consolidated financial statements	Correct metric and year
4	Which year had the highest free cash flow in the five-year period?	Cash-flow figures for FY2022–FY2026	Correct comparison across all five years
5	What was the operating margin in FY2026 and was it adjusted?	FY2026 report and definition/adjustment note	Preservation of the adjusted qualifier
6	What factors explain the change in free cash flow in FY2026?	Cash-flow statement, working-capital and capital expenditure notes	Avoid unsupported causal claims
7	What risks does the company disclose in the annual report?	Risk-management sections from relevant reports	Distinguish disclosed risk from interpretation

A useful evaluation log should record the question, expected answer, retrieved source page, actual answer, correctness, citation accuracy and notes. Tests should include both straightforward factual questions and difficult cases involving multiple years, table values, adjusted metrics and explanations that require more than one source passage.

Appendix D. Implementation Modules and Their Responsibilities

Module	Responsibility	Why it is separate
src/extract_reports.py	Extracts PDF text and preserves page markers	Allows extraction to be checked independently
src/rag.py	Loads/indexes text and performs semantic retrieval and response generation	Separates evidence retrieval from report writing
src/financial_data.py	Holds/organises the financial indicators used for comparison	Keeps key numeric series accessible for analysis
src/analyze_reports.py	Coordinates analyst tasks and generates the final analysis	Separates the multi-step workflow from retrieval
src/test_retrieval.py	Runs a retrieval check against the indexed corpus	Provides a repeatable way to test source retrieval

outputs/infosys_five_year_analysis.md	Human-readable financial analysis	Easy to inspect, edit and version-control
outputs/financial_trends.svg	Scalable financial trend visualisation	Can be embedded in documents and presentations
requirements.txt	Lists Python dependencies	Supports environment setup and reproducibility
README.md	Describes project setup and usage	Provides a starting point for reviewers

This modular structure makes it possible to improve one stage without rewriting the complete project. For example, OCR can be added to extraction, table-aware chunking can be added to indexing, or a larger evaluation dataset can be added without changing the report format.

Appendix E. Practical Business Recommendations

- Use the system first for exploratory questions and evidence discovery, then verify any number used in a formal presentation or decision.
- Keep a consistent definition for every financial indicator across reporting years; where the company uses an adjusted metric, preserve that label in charts and tables.
- Prioritise follow-up analysis where growth rates diverge, such as the difference between revenue growth and net-profit growth in FY2025.
- Investigate the FY2026 free-cash-flow decline by reviewing working capital, capital expenditure and operating cash-flow notes instead of assuming a cause from the trend alone.
- Build a small, verified benchmark of questions and source pages before claiming that the system is accurate across the entire report corpus.
- Retain a clear separation between the public source-code repository and a backup containing the large PDFs, extracted text and vector database. Confirm the backup can be opened before remote access ends.